

PathwayML-Ath: Discovering Novel Pathway-Like Gene Modules in *Arabidopsis thaliana* via GO Jaccard Coherence Features and Interpretable Machine Learning

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Abstract

Motivation: Curated pathway databases annotate fewer than 15% of protein-coding genes in *Arabidopsis thaliana*. Enrichment tools (ORA, GSEA) cannot identify novel functional modules absent from existing catalogues.

Results: We present PathwayML-Ath, a supervised framework that learns to distinguish curated pathway gene sets from non-pathway collections. Trained on 539 KEGG and AraCyc pathways with four types of Jaccard-calibrated hard negatives, the model encodes each gene set as a 91-dimensional vector of GO term frequencies (85), pairwise GO Jaccard coherence statistics (4), and size features (2). An XGBoost classifier achieves Repeated 5-fold CV AUROC of 0.949 ± 0.003 (SE). Ablation analysis reveals that Jaccard coherence statistics contribute the largest marginal gain (+0.357 AUROC when added to size-only baseline), while dense SVD/UMAP embeddings provide no additional benefit ($\Delta\text{AUROC} \leq 0.004$) under hard negatives—a methodological finding with implications for feature engineering in pathway-coherence classifiers. SHAP analysis identifies `jaccard_mean` and specific GO terms as top predictors. Applied to four test gene clusters, the model correctly scores a known ubiquitin proteolysis subset highly (C3: 0.825) while assigning low scores to incoherent mixtures, and identifies one candidate stress module (C1: 17 genes, score 0.841) not detected by ORA, supported by published CYP450/GST co-regulation literature.

Availability: Code, data, and reproducible notebook at <https://github.com/placeholder/pathway-ml-ath>.

Keywords: *Arabidopsis thaliana*; pathway discovery; XGBoost; SHAP; Gene Ontology; Jaccard coherence; hard negatives; KEGG; AraCyc

1 Introduction

Biological pathways encode the collective functional logic of the genome. For *Arabidopsis thaliana*, KEGG [Kanehisa et al., 2023] and AraCyc/PlantCyc [Hawkins et al., 2021] together annotate approximately 539 pathways, yet transcriptomic studies consistently identify co-regulated gene clusters unmapped to any existing entry [Obayashi et al., 2018, Rhee and Mutwil, 2014].

Overrepresentation analysis (ORA) and gene set enrichment analysis [GSEA; Subramanian et al., 2005] test enrichment within a predefined catalogue; a genuinely novel pathway will not be enriched for any existing annotation and will be missed. GSEA additionally requires a ranked gene list from

differential expression analysis, making it inapplicable to unranked gene sets. Unsupervised co-expression modules [Langfelder and Horvath, 2008] are pathway-agnostic but do not assess functional coherence directly.

Machine learning offers a complementary approach: if a classifier can learn intrinsic features distinguishing curated pathways from non-pathway gene collections, it can score any gene set for pathway-like coherence regardless of database membership. Related approaches exist for protein function prediction [Mostafavi et al., 2008] and drug-target interaction [Yamanishi et al., 2010], but no pathway-coherence classifier for plants has been reported.

We develop PathwayML-Ath with four key design principles. First, **Jaccard coherence features**: pairwise GO Jaccard similarity statistics quantifying internal functional cohesion, shown by ablation to be the single most important feature group. Second, **hard negative training**: four types of size-matched, Jaccard-calibrated negatives—including co-annotation clusters whose coherence matches real pathways—eliminating trivial classification shortcuts. Third, **embedding evaluation**: systematic comparison of linear (SVD) and non-linear (UMAP) embeddings, revealing that neither improves over hand-crafted Jaccard features under hard negatives. Fourth, **SHAP interpretability** [Lundberg and Lee, 2017] providing biologically meaningful explanations for each prediction.

2 Methods

2.1 Data sources

Pathway gene sets. KEGG *A. thaliana* pathway-gene associations were downloaded on 30 April 2026 via `rest.kegg.jp/link/ath/pathway`. AraCyc pathways were obtained from PlantCyc [Hawkins et al., 2021] (`aracyc_pathways.20251021`). Pathways with < 5 genes were excluded, yielding $N_{\text{path}} = 539$ (156 KEGG + 383 AraCyc). KEGG pathways have median 44 genes (range 5–2431); AraCyc pathways are smaller (median 12, range 5–102), reflecting specialised metabolite biosynthesis.

GO annotations. Gene Ontology annotations were obtained from the GO Consortium TAIR GAF file (`tair.gaf`, April 2026; Berardini et al. 2015): 27,435 genes with 6,743 GO terms. A dual-frequency filter retained 790 terms annotating 20–8,230 genes.

2.2 Feature engineering

Each gene set G is encoded as $\mathbf{x}(G) \in \mathbb{R}^{91}$ with three components.

GO term frequency profile ($d = 85$). After the dual-frequency filter (790 terms), a variance threshold (0.001) removes near-zero-variance terms. Remaining terms are ranked by mutual information (MI) with class labels. Rather than evaluating arbitrary k values, candidate k are derived from cumulative MI breakpoints at 30%, 50%, 70%, 80%, 90%, and 95% of total MI, plus the full set (Algorithm 1). The optimal $k = 85$ (at the 70% cumulative MI breakpoint) was selected by 5-fold CV AUROC. For each retained term t :

$$f_t(G) = \frac{|\{g \in G : t \in \text{GO}(g)\}|}{|G|} \quad (1)$$

Algorithm 1 Cumulative-MI-based GO feature selection

Require: GO frequency matrix $\mathbf{F}$, labels $\mathbf{y}$, MI fractions $\mathcal{Q} = \{0.3, 0.5, 0.7, 0.8, 0.9, 0.95\}$

- 1: Remove features with variance < 0.001
 - 2: Compute MI scores: $\text{MI}_j = I(F_j; y)$ for each feature j
 - 3: Rank features by MI; compute cumulative MI fraction C_k
 - 4: **for** each $q \in \mathcal{Q}$ **do**: $k_q \leftarrow \min\{k : C_k \geq q\}$
 - 5: **end for**
 - 6: $k^* \leftarrow \arg \max_{k \in \{k_q\} \cup \{p'\}} \text{CV-AUROC}(k)$
 - 7: **return** top k^* features
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Pairwise GO Jaccard statistics ($d = 4$). For pairs (g_i, g_j) in G (subsampled to ≤ 15 genes, **sorted()** for determinism):

$$J(g_i, g_j) = \frac{|\text{GO}(g_i) \cap \text{GO}(g_j)|}{|\text{GO}(g_i) \cup \text{GO}(g_j)|} \quad (2)$$

Four statistics—mean, standard deviation, minimum, and maximum—quantify internal functional coherence.

Size features ($d = 2$). $|G|$ and $\log(1 + |G|)$.

Why not embeddings? We evaluated TF-IDF/SVD (linear, $d = 20$) and UMAP (non-linear, $d = 20$; applied to SVD-50 pre-reduced data) embeddings of the gene-level GO matrix. Each gene set’s embedding is the element-wise mean of its members’ vectors. Under hard negatives, neither improved over Jaccard + Size ($\Delta\text{AUROC} \leq 0.004$; Table 2). Embeddings capture a gene set’s centroid in GO space, but pathway discrimination requires the *pairwise similarity structure*—the distribution of $J(g_i, g_j)$ values—which centroid-based representations cannot capture. We therefore exclude embeddings from the final model, reducing D from 111 to 91 without performance loss.

2.3 Training data construction

Positive samples: 539 curated pathways ($y = 1$). Negative samples: $n^- = 1,078$ ($y = 0$, ratio 1:2), four types in equal proportions (25% each), all with sizes sampled from the real pathway size distribution:

1. **Jaccard-matched random** ($n = 270$): gene sets filtered by rejection sampling to have internal Jaccard $\geq P_{25}$ of real pathways, preventing the model from using coherence level as a trivial shortcut.
2. **Co-annotation clusters** ($n = 270$): seeded from a random gene, expanded with GO-similar neighbours (50% coherent, 50% noise). The noise fraction (0.5) was calibrated by grid search over $\{0.3, 0.4, 0.5, 0.6, 0.7\}$ to match the real pathway Jaccard distribution (median ≈ 0.13). These test whether the model can distinguish genuine pathway topology from mere GO co-annotation.
3. **Cross-pathway chimeras** ($n = 270$): equal subsets from 2–3 different pathways, creating gene sets with moderate but heterogeneous coherence.
4. **Shuffled pathways** ($n = 268$): real pathways with 40–60% of genes replaced by random genes, testing sensitivity to disruption.

Equal proportions (25% each) avoid biasing the decision boundary toward any single negative type. Size-only AUROC = 0.461 (below chance) confirms size is not a shortcut.

2.4 Model training and evaluation

An XGBoost classifier [Chen and Guestrin, 2016] (v3.2; `n_estimators=500`, `max_depth=5`, `learning_rate=0.03`, `scale_pos_weight=2`) was trained. Two baselines: Random Forest [Breiman, 2001] and Logistic Regression. Model selection used Repeated 5-fold CV ($\times 5$ repeats, 25 folds total; standard error reported). Held-out test set: 20% stratified.

2.5 Ablation study design

Three ablation types: (i) single-group removal (3 configs); (ii) pairwise removal (all $C(3, 2) = 3$ configs); (iii) cumulative addition (Size $\rightarrow$ +Jaccard $\rightarrow$ +GO freq). Additionally, SVD and UMAP embeddings were evaluated as add-on features.

2.6 SHAP interpretability

TreeExplainer [Lundberg and Lee, 2017] provides per-feature SHAP values ϕ_i (log-odds). Global importance: mean $|\phi_i|$ across test samples.

2.7 Candidate scoring and ORA comparison

Four gene clusters were scored: C1 (17 oxidative stress genes from no single pathway), C2 (11 photosynthesis genes), C3 (14 ubiquitin proteolysis genes), C4 (mixed pathway + random). Hypergeometric ORA with Bonferroni correction was applied for comparison.

3 Results

3.1 Classifier performance

Table 1 summarises performance. XGBoost achieves Repeated CV AUROC 0.949 ± 0.003 (SE) and test AUROC 0.941, substantially exceeding Logistic Regression (0.879).

Table 1: Classification performance. CV: Repeated 5-fold $\times 5$ (25 folds; $\pm$ SE). Test: held-out 20%.

Model	CV AUROC (mean $\pm$ SE)	Test AUROC
XGBoost	0.949 ± 0.003	0.941
Random Forest	0.949 ± 0.002	0.922
Logistic Regression	0.911 ± 0.003	0.879
Random baseline	0.500	0.500

3.2 Ablation study

Table 2 presents the complete ablation. Three key findings emerge.

First, the cumulative analysis reveals the contribution hierarchy: Jaccard provides the largest marginal gain (+0.357 when added to Size-only), GO frequency adds +0.123, and the full model reaches 0.941.

Second, Jaccard is the only feature group whose removal causes a large drop (-0.049); GO frequency removal has a smaller effect (-0.013), and Size removal is moderate (-0.023).

Third, adding SVD ($\Delta = -0.003$) or UMAP ($\Delta = -0.002$) embeddings to the full model provides no benefit, confirming that embeddings are redundant under hard negatives.

Table 2: Ablation study. Three feature groups: GO freq ($d = 85$), Jaccard ($d = 4$), Size ($d = 2$). Δ : difference from full model test AUROC.

Configuration	d	Test AUROC	Δ
<i>Single removal</i>			
Full model	91	0.941	—
−GO freq	6	0.928	−0.013
−Jaccard	87	0.892	−0.049
−Size	89	0.918	−0.023
<i>Pairwise removal</i>			
−GO −Jaccard (= Size only)	2	0.461	−0.480
−GO −Size (= Jaccard only)	4	0.769	−0.172
−Jaccard −Size (= GO only)	85	0.871	−0.070
<i>Cumulative addition (marginal Δ)</i>			
Size only	2	0.461	—
+ Jaccard	6	0.818	+0.357
+ GO freq (= Full)	91	0.941	+0.123
<i>Embedding add-ons</i>			
Full + SVD (linear, $d = 20$)	111	0.938	−0.003
Full + UMAP (non-linear, $d = 20$)	111	0.939	−0.002

3.3 SHAP feature attribution

SHAP analysis confirms the ablation: `jaccard_mean` and `jaccard_std` rank among the top 5 features globally. Specific GO terms (membrane, protein binding, carbohydrate metabolism) contribute additional signal, consistent with pathway genes sharing functional annotations beyond raw pairwise similarity. Per-candidate waterfall plots (supplementary) trace each prediction to individual features.

3.4 Candidate pathway evaluation

Table 3: Candidate evaluation. ORA: Bonferroni $p < 0.05$. Novel: score ≥ 0.50 and max Jaccard < 0.40 .

Cand.	n	Score	Max J	ORA	Novel?
C1 (stress module)	17	0.841	0.000	No	✓
C2 (photosynthesis)	11	0.019	0.143	Yes	×
C3 (ubiquitin prot.)	14	0.825	0.086	Yes	×
C4 (mixed)	9	0.016	0.027	Yes	×

C3 (positive control): 14 ubiquitin proteolysis genes correctly receive score 0.825, confirmed by ORA ($p_{\text{adj}} < 10^{-30}$).

C1 (novel candidate): 17 genes annotated with oxidative stress (GO:0006979), monooxygenase/CYP450 activity (GO:0004497), and glutathione metabolism (GO:0006749), belonging to no single KEGG or AraCyc pathway (max Jaccard = 0.000). The model assigns score 0.841, yet ORA finds no significant enrichment—precisely the use case PathwayML-Ath is designed for. This CYP450/GST coordination module is consistent with combined-stress transcriptomic responses reported by Zandalinas et al. [2020] and the ROS signalling networks reviewed by Mittler et al. [2022].

C2 (photosynthesis subset, score 0.019) demonstrates that even within a single pathway, gene subsets can have low internal GO coherence—the model evaluates multi-dimensional coherence, not pathway membership. C4 (mixed, score 0.016) correctly receives a low score.

4 Discussion

Jaccard coherence is the primary pathway signal. The cumulative ablation shows that four Jaccard statistics capture the core pathway signal: the jump from 0.461 to 0.818 demonstrates that pairwise functional similarity is the defining property of pathway gene sets. This is biologically intuitive—pathway members catalyse sequential or parallel reactions and share GO annotations for the relevant metabolic or signalling process.

Embeddings are redundant under hard negatives. Neither SVD nor UMAP embeddings improve performance when co-annotation cluster negatives are used. We attribute this to a fundamental representational limitation: embeddings capture a gene set’s centroid in GO space, but pathway discrimination requires the *distribution* of pairwise similarities. Co-annotation clusters have similar centroids to real pathways (by construction) but lack the uniform pairwise coherence characteristic of true pathways. Jaccard statistics capture this distribution; embedding centroids cannot. This finding cautions against evaluating embedding-based features only against easy negatives, where their advantage may be artifactual.

Complementarity with ORA. C1 demonstrates that PathwayML-Ath detects coherent modules invisible to ORA. The recommended workflow: (a) ORA/GSEA for known-pathway enrichment; (b) PathwayML-Ath for residual clusters; (c) SHAP for biological interpretation.

Reproducibility. Two sources of non-determinism were identified and fixed: (i) global random seed consumption order (addressed by executing all cells sequentially); (ii) `set()` to `list()` ordering in Jaccard computation (fixed by using `sorted()`). Repeated CV ($5 \times 5 = 25$ folds) provides stable AUROC estimates ($\text{SE} = 0.003$).

Limitations. (i) GO annotation incompleteness ($\sim 30\%$ of genes lack GO); protein language model embeddings [Lin et al., 2023] could address this. (ii) Model reflects specific database versions (KEGG April 2026, AraCyc October 2025). (iii) C1 requires experimental validation. (iv) Leave-one-pathway-family-out evaluation would further test generalisation across metabolic categories. (v) Non-linear embeddings beyond UMAP (e.g., gene2vec [Du et al., 2019] on co-expression data, Node2Vec on GO DAG) remain to be evaluated.

5 Conclusion

PathwayML-Ath demonstrates that four pairwise GO Jaccard statistics, augmented by GO term frequencies, are sufficient to discriminate curated pathway gene sets from challenging hard negatives (Repeated CV AUROC 0.949 ± 0.003). Dense embeddings (SVD, UMAP) are redundant under Jaccard-calibrated negatives. The framework identifies a CYP450/GST stress module invisible to ORA, providing a prioritised target for experimental validation. All code and a reproducible notebook are released with the manuscript.

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Data Availability

KEGG: <https://rest.kegg.jp>; GO: <https://current.geneontology.org/annotations/tair.gaf.gz>; AraCyc: PlantCyc; Code: <https://github.com/placeholder/pathway-ml-ath>.

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